



Syntax IV

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Please pick up your **name card** from the front

Previously on 201

Phrase Structure Rules (PSRs) for NP, PP, AdjP and AdvP

- **NP** $\rightarrow$ (D) (AdjP+) **N** (PP+)
- **PP** $\rightarrow$ **P** NP
- **AdjP** $\rightarrow$ (AdvP) **Adj**
- **AdvP** $\rightarrow$ (AdvP) **Adv**

Exercise I

NP, **AdjP**, **AdvP**, **PP**

draw a **syntax tree** for the underlined phrases

- I ate an apple.
- John went into the building.
- They're like peas in a pod.
- The bakery sells wonderfully fresh bread.
- I like to drive really slowly.
- Mary helped the man in the car with the flat tire.
- I threw a ball to the dog with a collar in the park.

NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

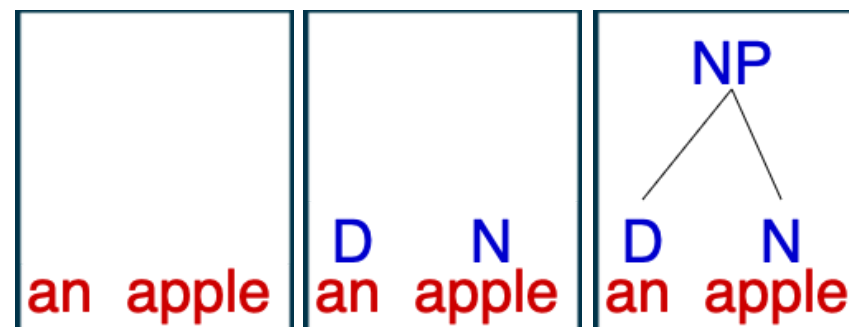
AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → NP (**T**) VP

CP → **C** TP

VP → (AdvP+) **V** (NP) (NP) (AdvP+) (PP+) (AdvP+)



- in **f** does with the flat tire modify 'the man' or 'the car'?
- what about in the park in **g**?

Exercise I

NP, AdjP, AdvP, PP

NP → (D) (AdjP+) N (PP+)

PP → P NP

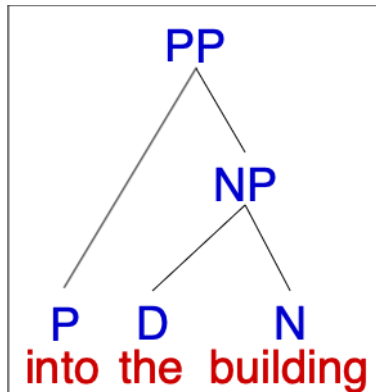
AdjP → (AdvP) Adj

AdvP → (AdvP) Adv

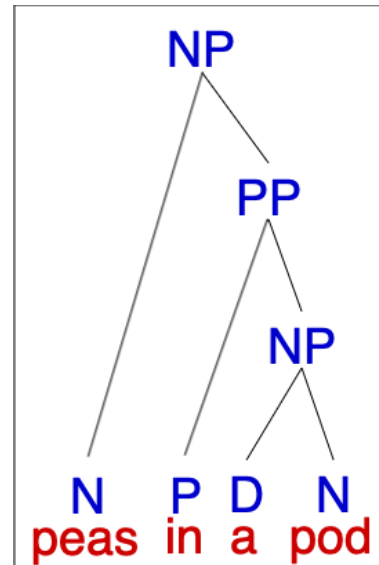
TP → NP (T) VP

CP → C TP

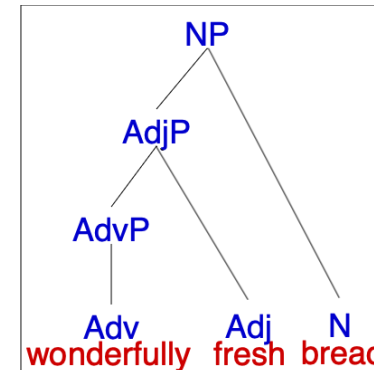
VP → (AdvP+) V (NP) (NP) (AdvP+) (PP+) (AdvP+)



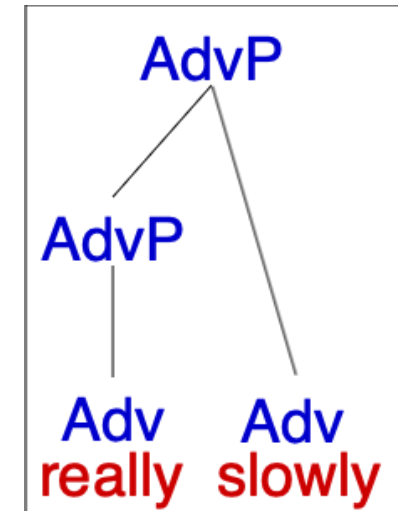
b. 'into the building'



c. 'peas in a pod'



d. 'wonderfully fresh bread'

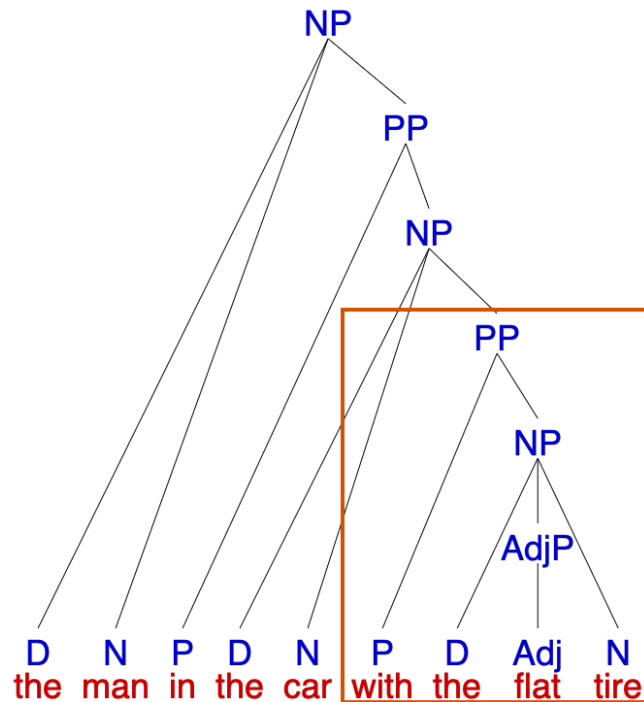


e. 'really slowly'

Exercise I

NP, AdjP, AdvP, PP

f. the man in the car with the flat tire



NP → (D) (AdjP+) N (PP+)

PP → P NP

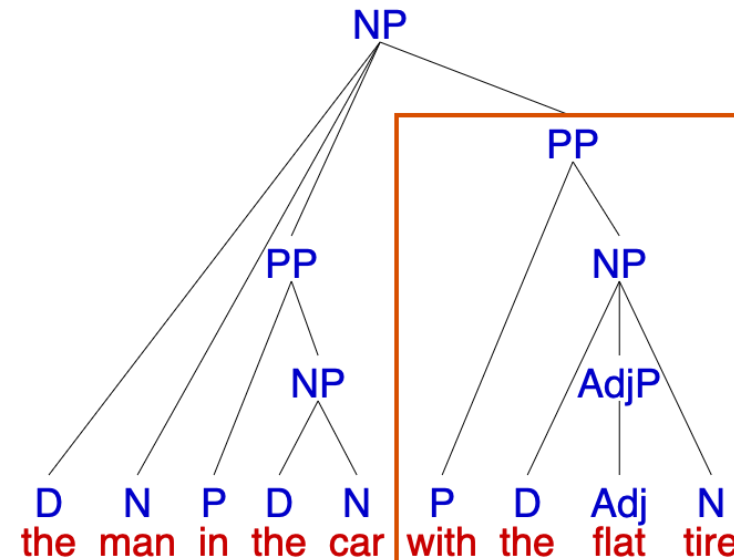
AdjP → (AdvP) Adj

AdvP → (AdvP) Adv

TP → NP (T) VP

CP → C TP

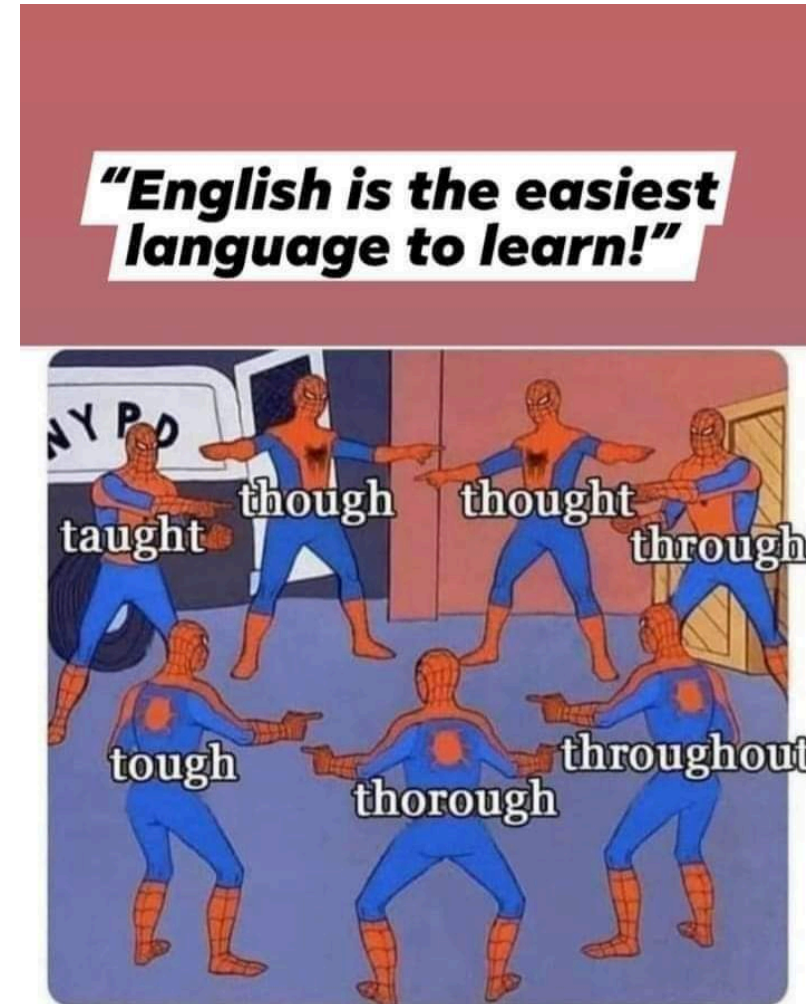
VP → (AdvP+) V (NP) (NP) (AdvP+) (PP+) (AdvP+)



Verb Phrases (VP)

- a verb phrase (**VP**) consists of **a verb** and everything that **modify** it
- generally, this includes **everything** in a sentence except the **subject** (the entity that performs the action indicated by the verb)
- the simplest **VP** is a single verb:
 - *Arthur ate.* **VP** → V
 - *Merlin left.* **VP** → V

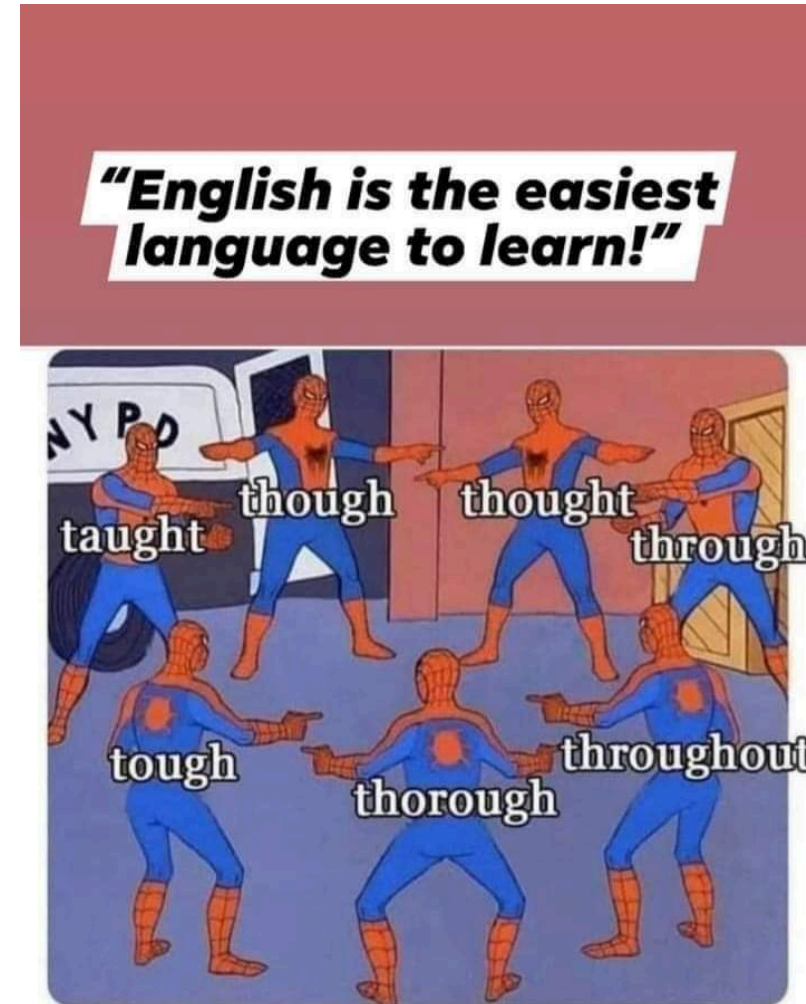
VP → V



Verb Phrases (VP)

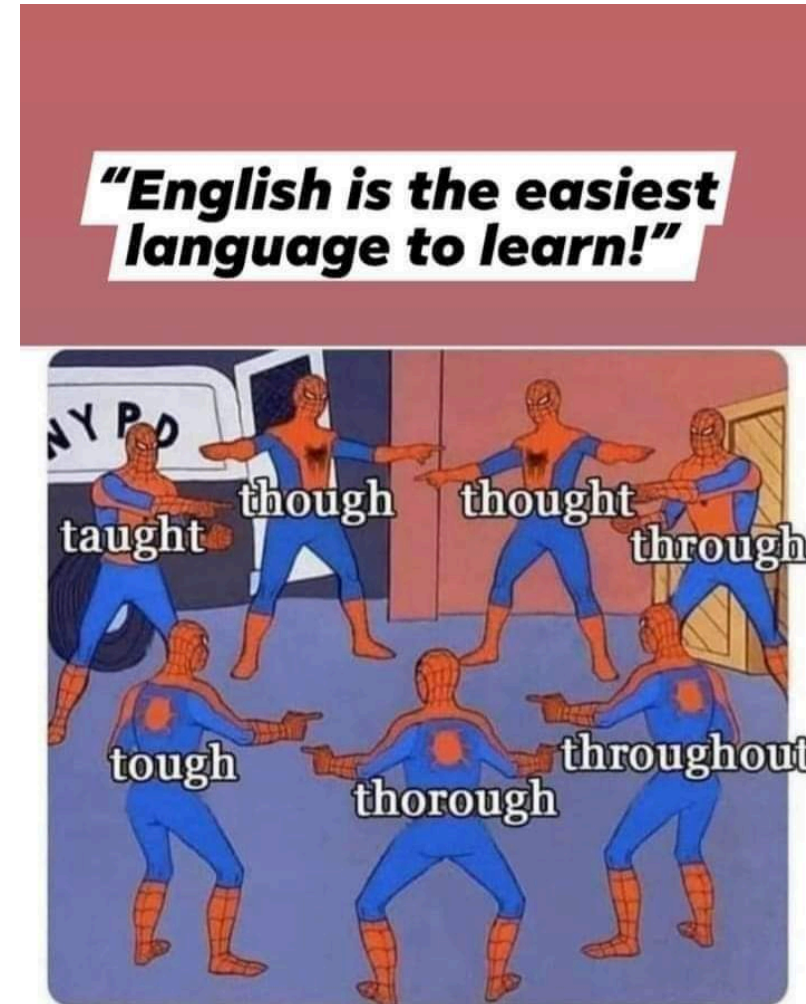
- besides the verb, **VP**s also contain any **objects** of the verb
- **direct object**: person or entity **affected** by the action indicated by the verb
direct objects are **NPs**:
 - Peter baked [a cake].
- **indirect object**: person or entity that **receives** something or **benefits** from the action
indirect object can be **PPs** or **NPs**:
 - Peter baked [a cake] [for his mom].
 - Peter baked [his mom] [a cake].

VP → V (NP) (NP/PP)



Verb Phrases (VP)

- in addition to objects, a **VP** can contain **adjuncts**
- they are **optional** elements that describe the **action** and therefore **modify** the verb
- adjuncts may be **PPs** or **AdvPs**:
 - The train arrived [at Shanghai].
VP → V **PP**
 - The train arrived [quickly].
VP → V **AdvP**
- a sentence can have multiple **adjuncts**:
 - The train arrived [quickly] [at Shanghai].



VP → **(AdvP+)** **V** **(NP)** **(NP)** **(AdvP+)** **(PP+)**
(AdvP+)

just remember a few things:

1. a **VP** always has a verb **head**
2. a **VP** can contain many **optional** things like **NP**, **PP**, and **AdvP** arguments and adjuncts, **BUT**:
3. a **VP** does NOT contain the subject of the verb
4. **VPs** will NOT directly contain an **AdjP** (since AdjPs describe nouns, not verbs)
5. any **NPs** or **PPs** will always go after the verb
6. objects and adjuncts will always be **siblings** of the verb

Tense Phrases (TP, Sentence)

sentences are also known as **tense phrases** (TP, or occasionally S), and it consists of:

- **NP** – a required subject
- **T** – an optional slot called "tense" that can be filled by:
 1. modal verbs (*can, might, should...*)
 2. auxiliary verbs (*is, are, has, will...*)
 3. the infinitive marker *to* as in *to go, to see*
 4. PAST and PRESENT if there is no overt tense-marker as above
- **VP** – a required verb phrase (sometimes called the predicate)

		Simple	Continuous	Perfect	Perfect Continuous
skyeng	Past	 I lay down on the couch	 I was lying on the couch	 I had already lain down, when my human called me for lunch	 I had been lying on the couch for 2 hours, when a TV series started
	Present	 I usually lie on the couch all day long	 Now I'm lying on the couch and watching a TV series	 I have never lain on the couch for so long	 I have been lying on the couch for 2 hours already
	Future	 I will never get up off the couch	 Tomorrow I will be lying on the couch all day long	 I will have gained 365 days of couch potato experience by the next summer	 I will have been lying on the couch for 1 year by 2021

Steps for Drawing Trees

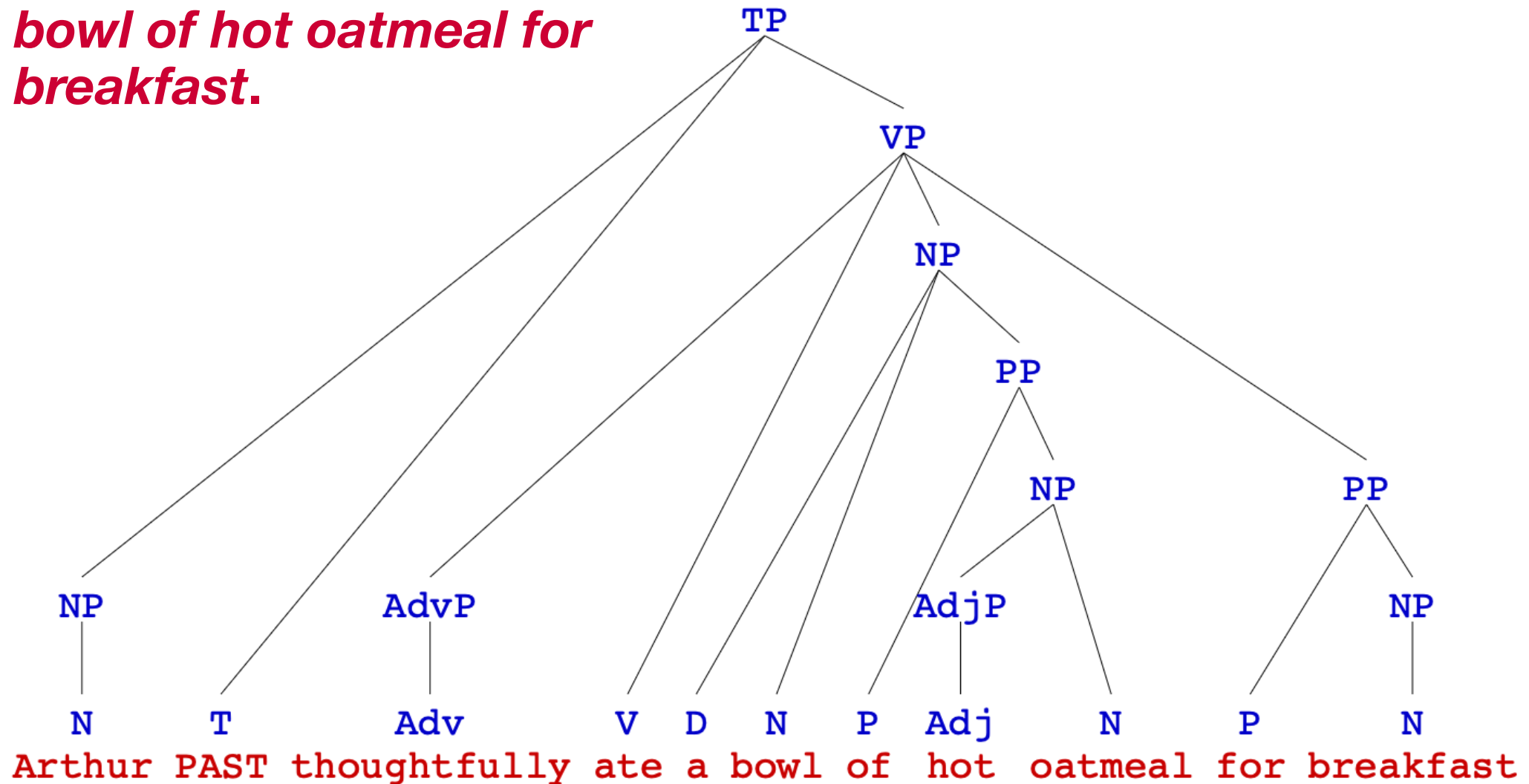
1. write the sentence at the bottom of the page – leave yourself **plenty of room**
2. label the **part of speech** of each word
3. sentences have a subject **NP** and predicate **VP** – try to identify these first, and draw the subject
4. trees should ALWAYS follow **PSRs**, so try to identify elements that can be part of rules for different phrases
5. every **noun**, **adjective**, **adverb**, **preposition** has its own corresponding phrase **NP**, **AdjP**, **AdvP**, **PP**
6. smaller trees first, then figure out how they connect to the rest of the tree

some notes

- VPs can have lots of elements, so figure them out last
- **pay attention to what modifies what** – any phrase that modifies a word should be its **sibling**
- because of the way English is structured, it can often be easier to start at the end and work **backwards**

Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

let's do a this tree **bottom-to-top** together



Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

first, write out the **words**
and label the parts of speech

NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → NP (**T**) VP

CP → **C** TP

VP → (AdvP+) **V** (NP) (NP)
(AdvP+) (PP+) (AdvP+)

Arthur thoughtfully ate a bowl of hot oatmeal for breakfast

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first, write out the words
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NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

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AdvP → (AdvP) **Adv**

TP → NP (**T**) VP

CP → **C** TP

VP → (AdvP+) **V** (NP) (NP)
(AdvP+) (PP+) (AdvP+)

N		Adv		V	D	N	P	Adj		N	P		N
Arthur		thoughtfully		ate	a	bowl	of	hot		oatmeal	for		breakfast

Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

find out where **T** should be
since it breaks apart **NP** and **VP** in the sentence

usually we look for the **subject** and the **main verb**
in this case, it is *Arthur* and *ate*

NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

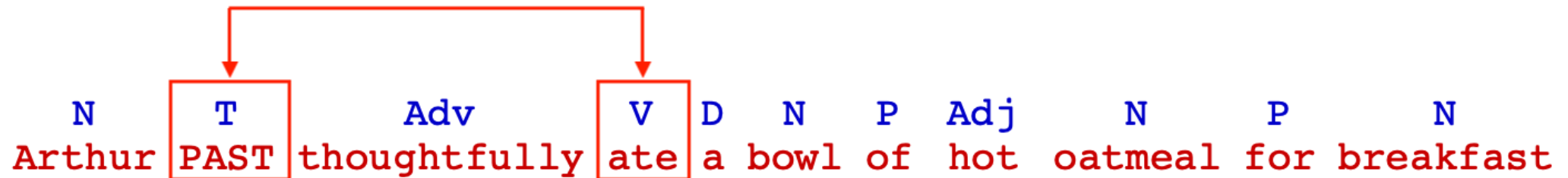
AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → NP (**T**) VP

CP → **C** TP

VP → (AdvP+) **V** (NP) (NP)
(AdvP+) (PP+) (AdvP+)



Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

draw a tree for the whole sentence by using **TP** →
NP **T** **VP** at the top

identify any **T** and connect it

if there's no **T**, you might need to think about
if it is **PRESENT** or **PAST** tense

NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

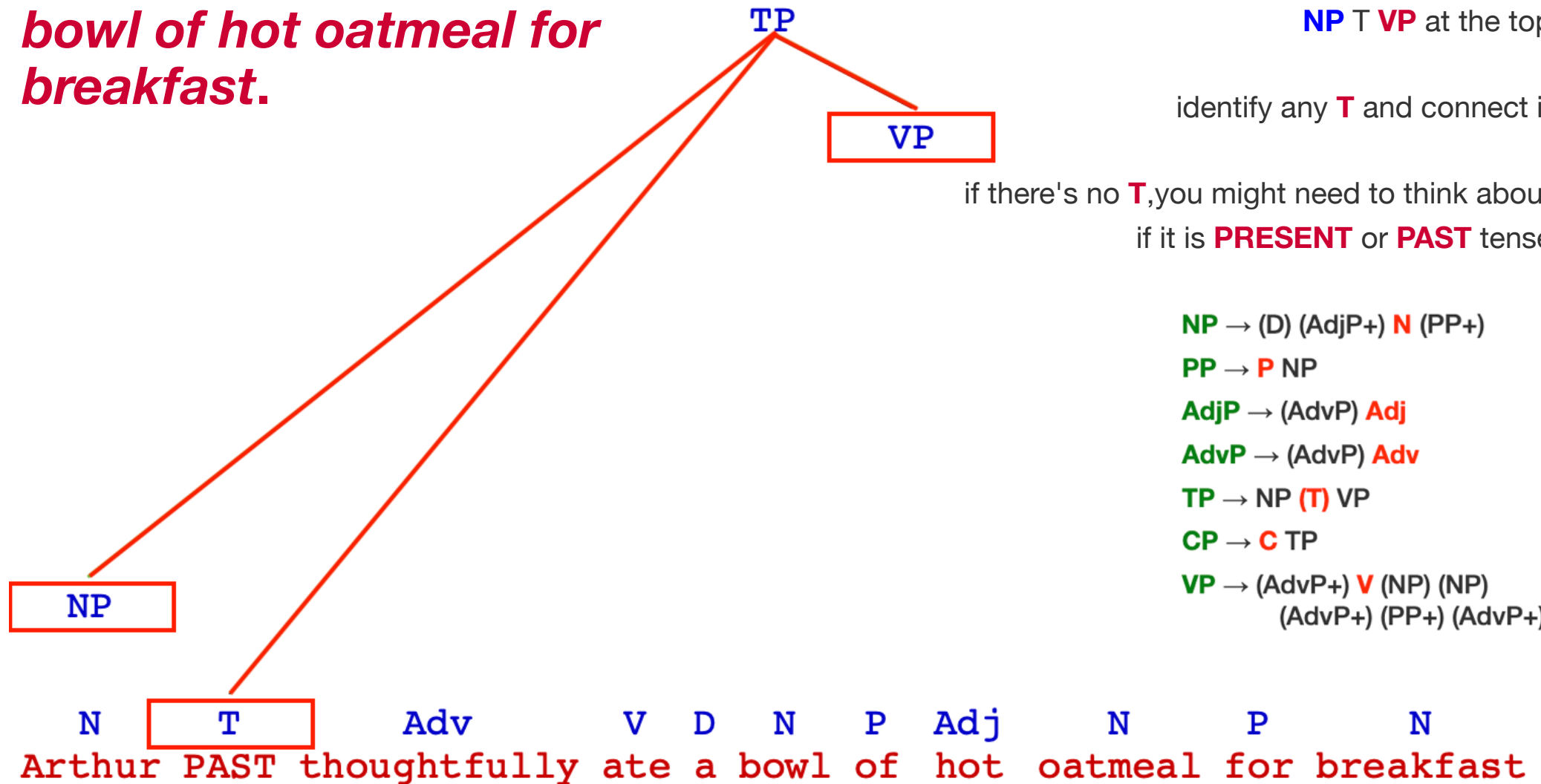
AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → NP (**T**) VP

CP → **C** TP

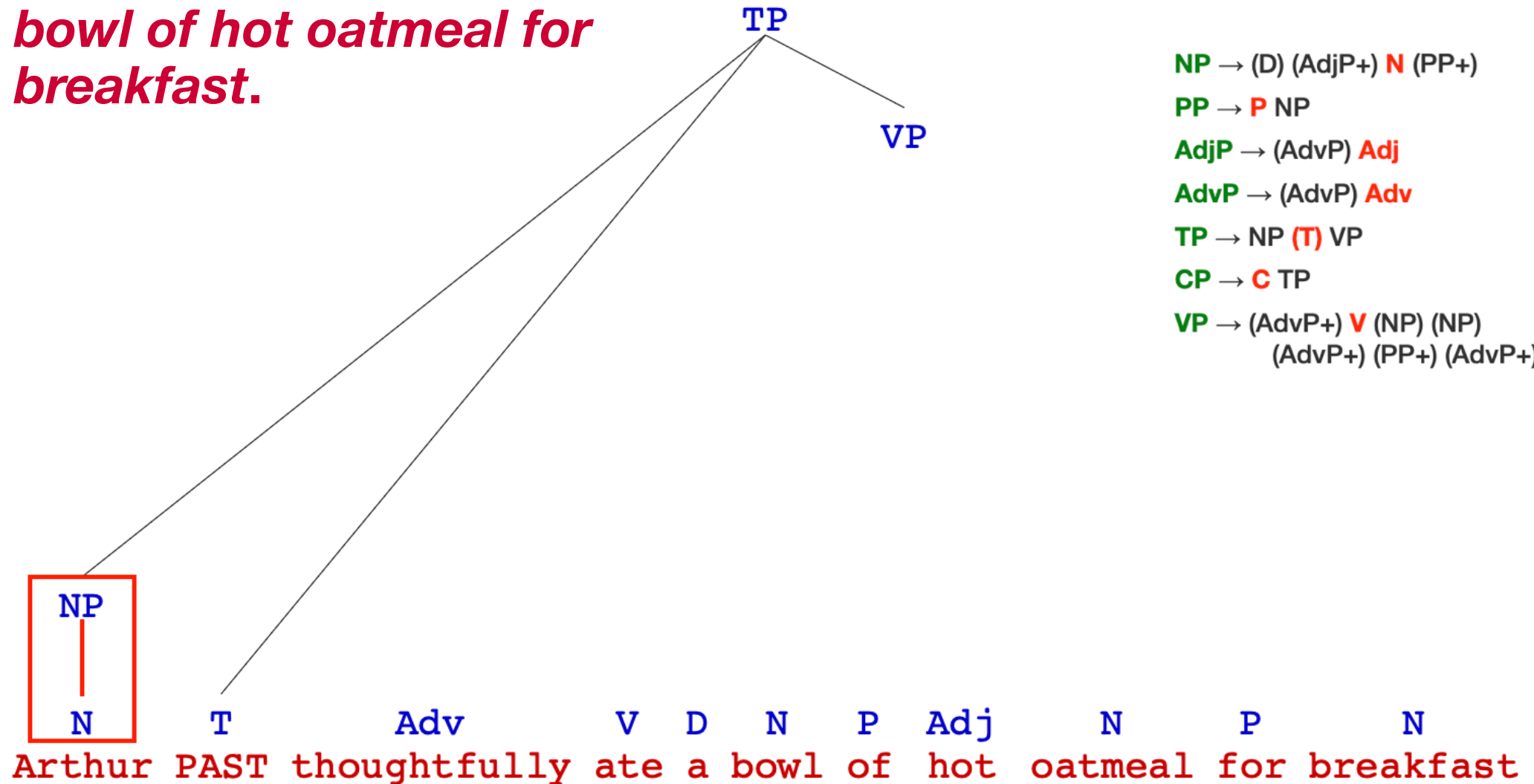
VP → (AdvP+) **V** (NP) (NP)
(AdvP+) (PP+) (AdvP+)



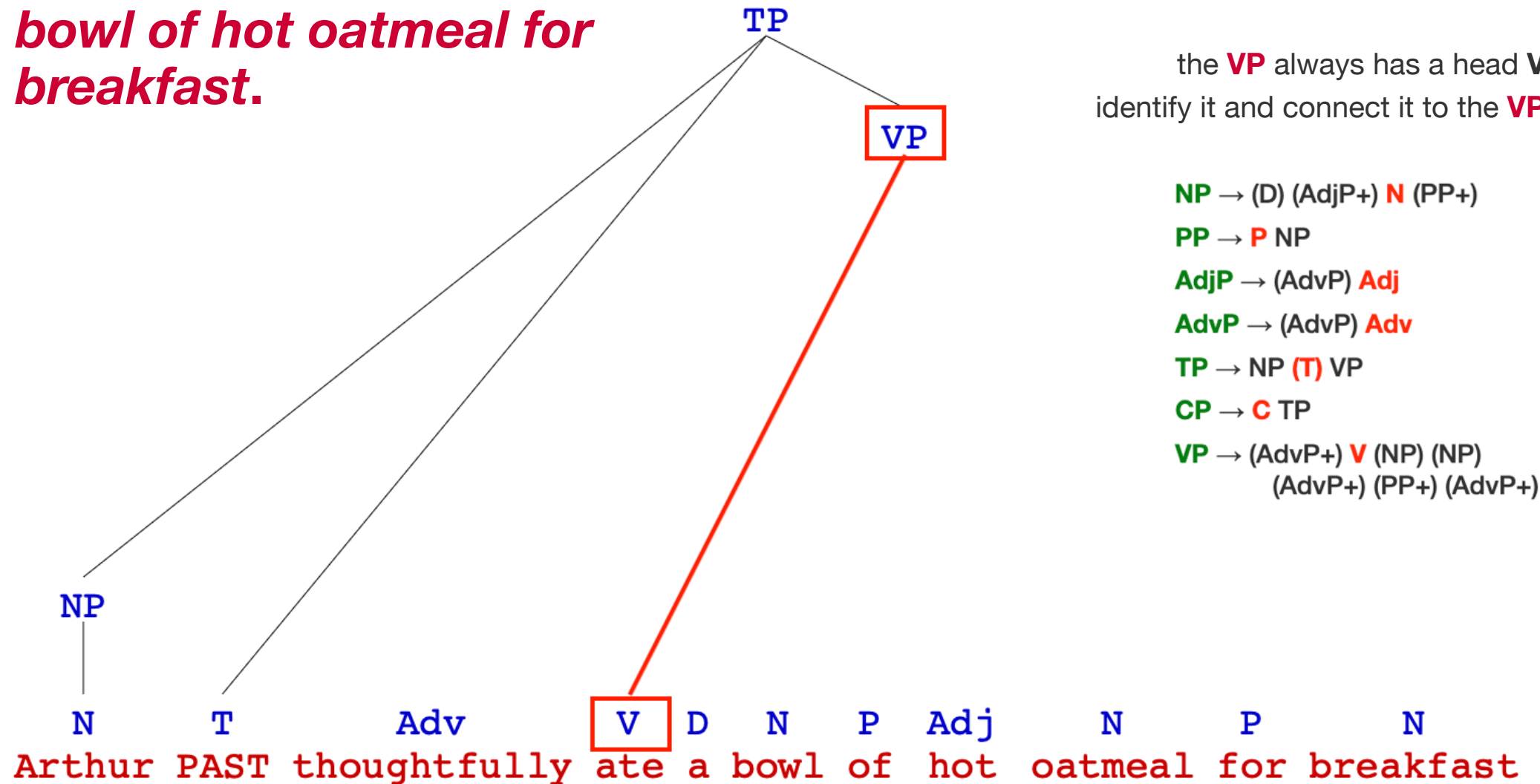
Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

identify the subject **NP** and draw its tree

- NP** → (D) (AdjP+) **N** (PP+)
- PP** → **P** NP
- AdjP** → (AdvP) **Adj**
- AdvP** → (AdvP) **Adv**
- TP** → NP (**T**) VP
- CP** → **C** TP
- VP** → (AdvP+) **V** (NP) (NP)
(AdvP+) (PP+) (AdvP+)



Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.



everything left over is the **VP**

the **VP** always has a head **V**
identify it and connect it to the **VP**

NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → NP (**T**) VP

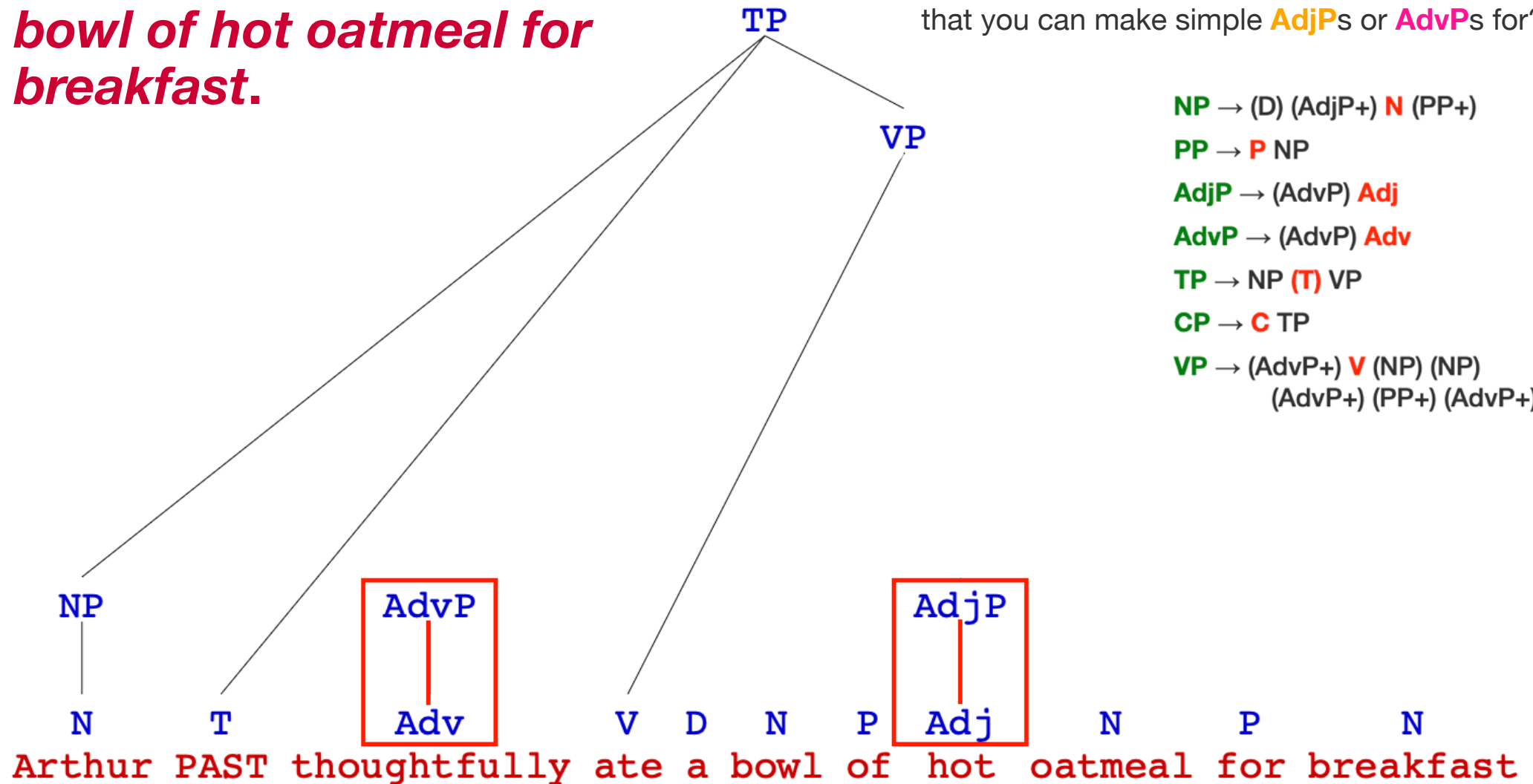
CP → **C** TP

VP → (AdvP+) **V** (NP) (NP)
(AdvP+) (PP+) (AdvP+)

Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

check there are any adjectives or adverbs that you can make simple **AdjPs** or **AdvPs** for?

- NP** → (D) (AdjP+) **N** (PP+)
- PP** → **P** NP
- AdjP** → (AdvP) **Adj**
- AdvP** → (AdvP) **Adv**
- TP** → NP (**T**) VP
- CP** → **C** TP
- VP** → (AdvP+) **V** (NP) (NP)
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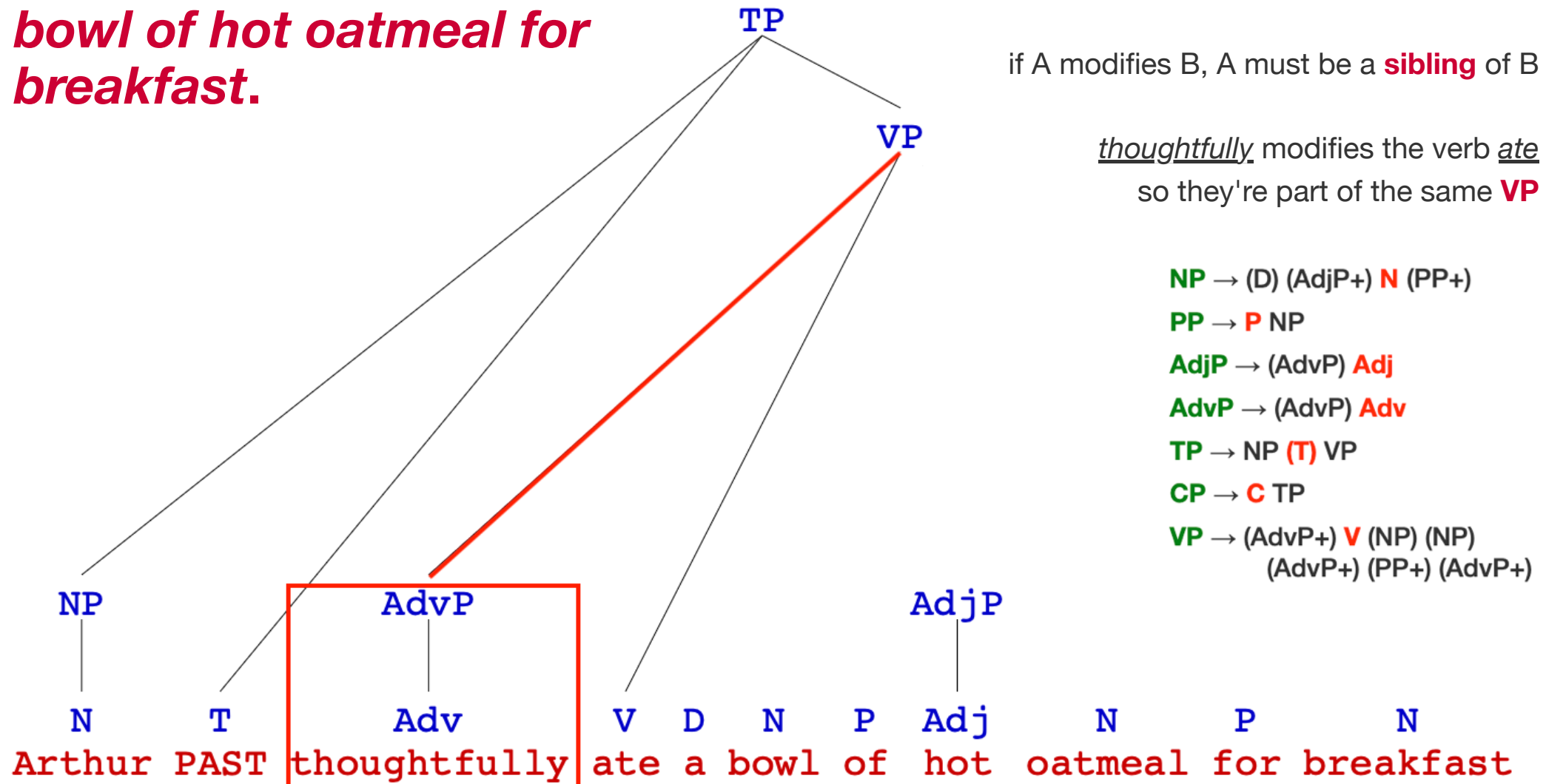


Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

what does this **AdvP** modify?

if A modifies B, A must be a **sibling** of B

thoughtfully modifies the verb ate
so they're part of the same **VP**

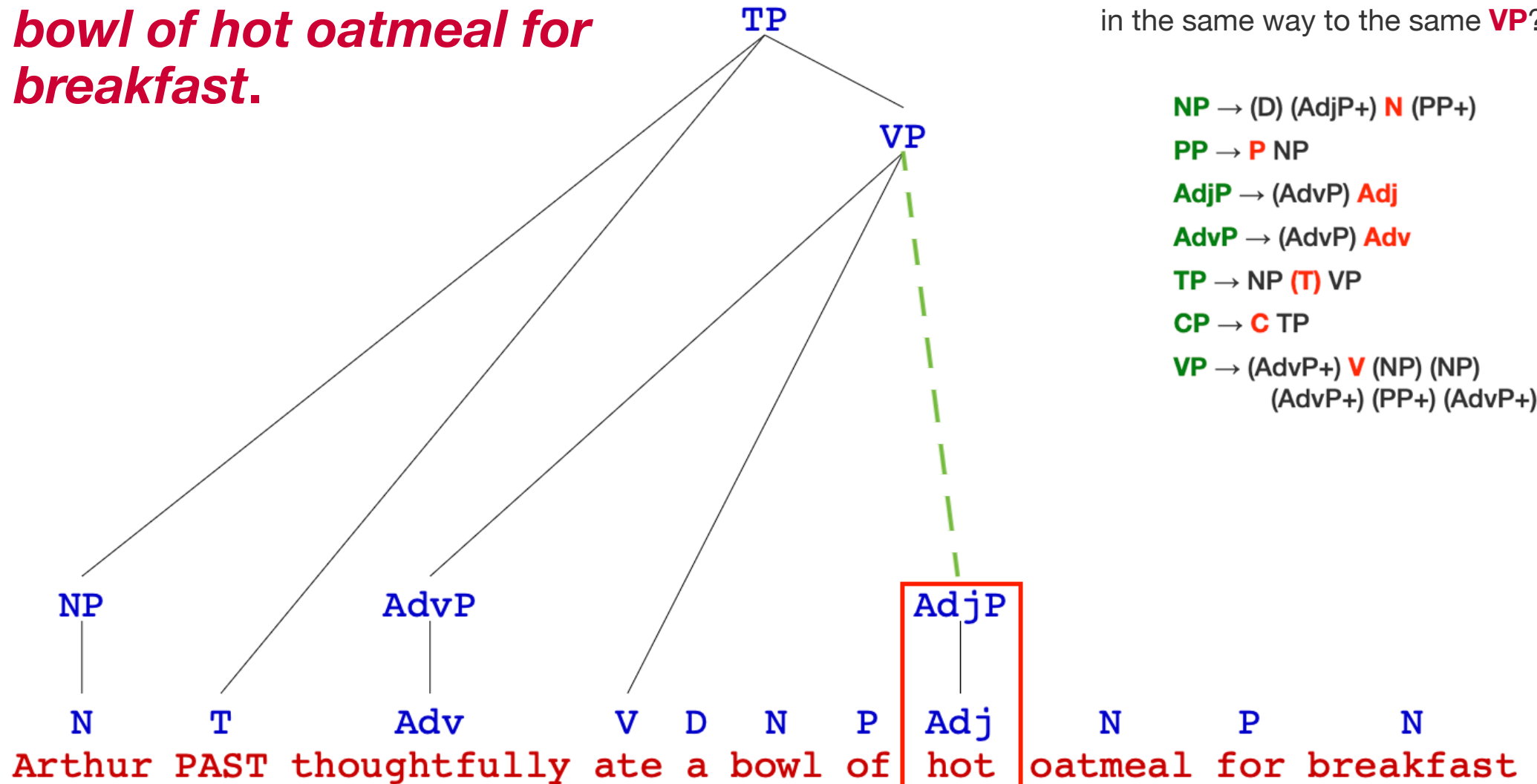


- NP → (D) (AdjP+) **N** (PP+)
- PP → **P** NP
- AdjP → (AdvP) **Adj**
- AdvP → (AdvP) **Adv**
- TP → NP (**T**) VP
- CP → **C** TP
- VP → (AdvP+) **V** (NP) (NP)
(AdvP+) (PP+) (AdvP+)

Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

can we connect this **AdjP**
in the same way to the same **VP**?

- NP** → (D) (AdjP+) **N** (PP+)
- PP** → **P** NP
- AdjP** → (AdvP) **Adj**
- AdvP** → (AdvP) **Adv**
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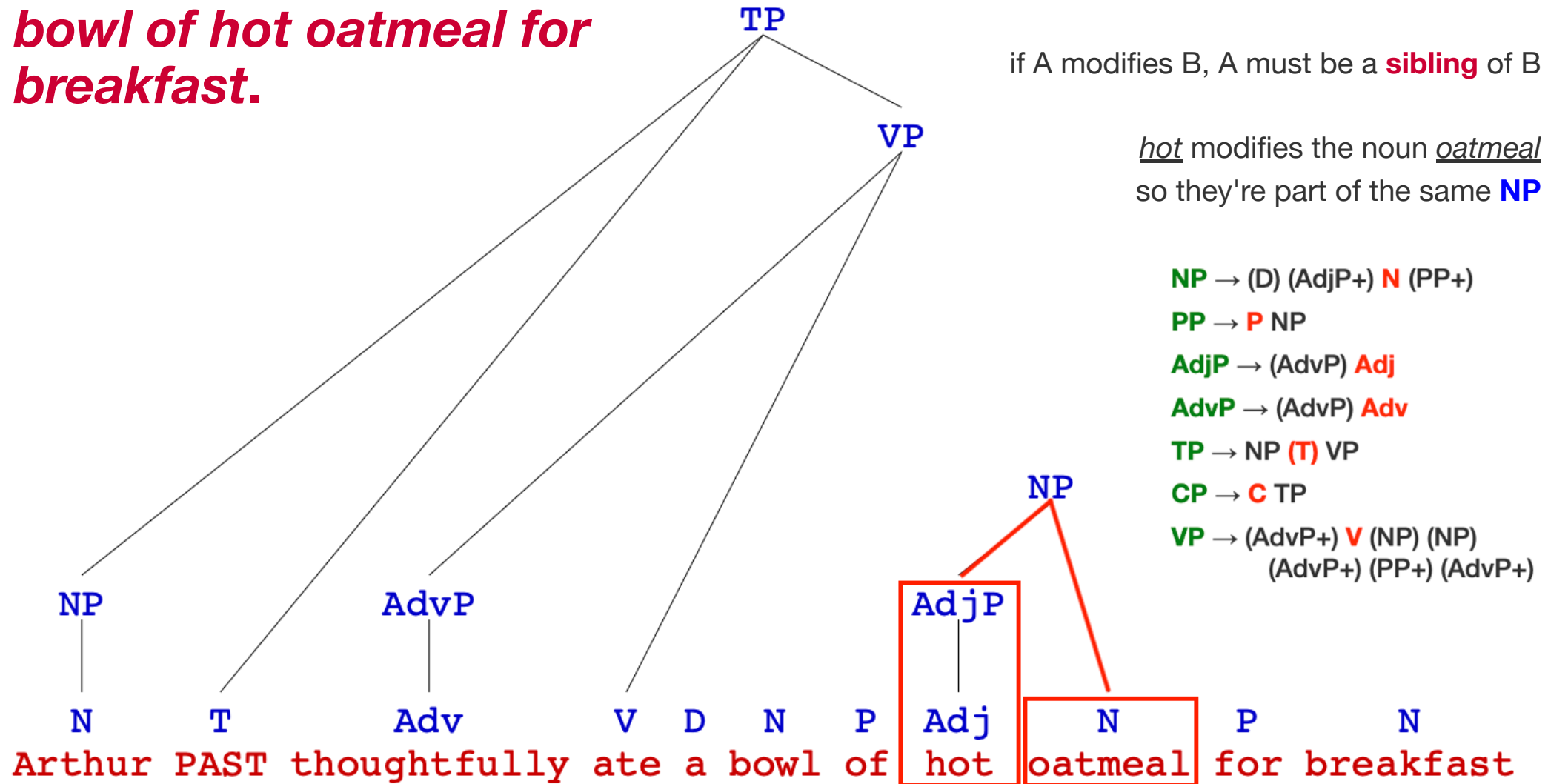


Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

what does this **AdjP** modify?

if A modifies B, A must be a **sibling** of B

hot modifies the noun oatmeal
so they're part of the same NP

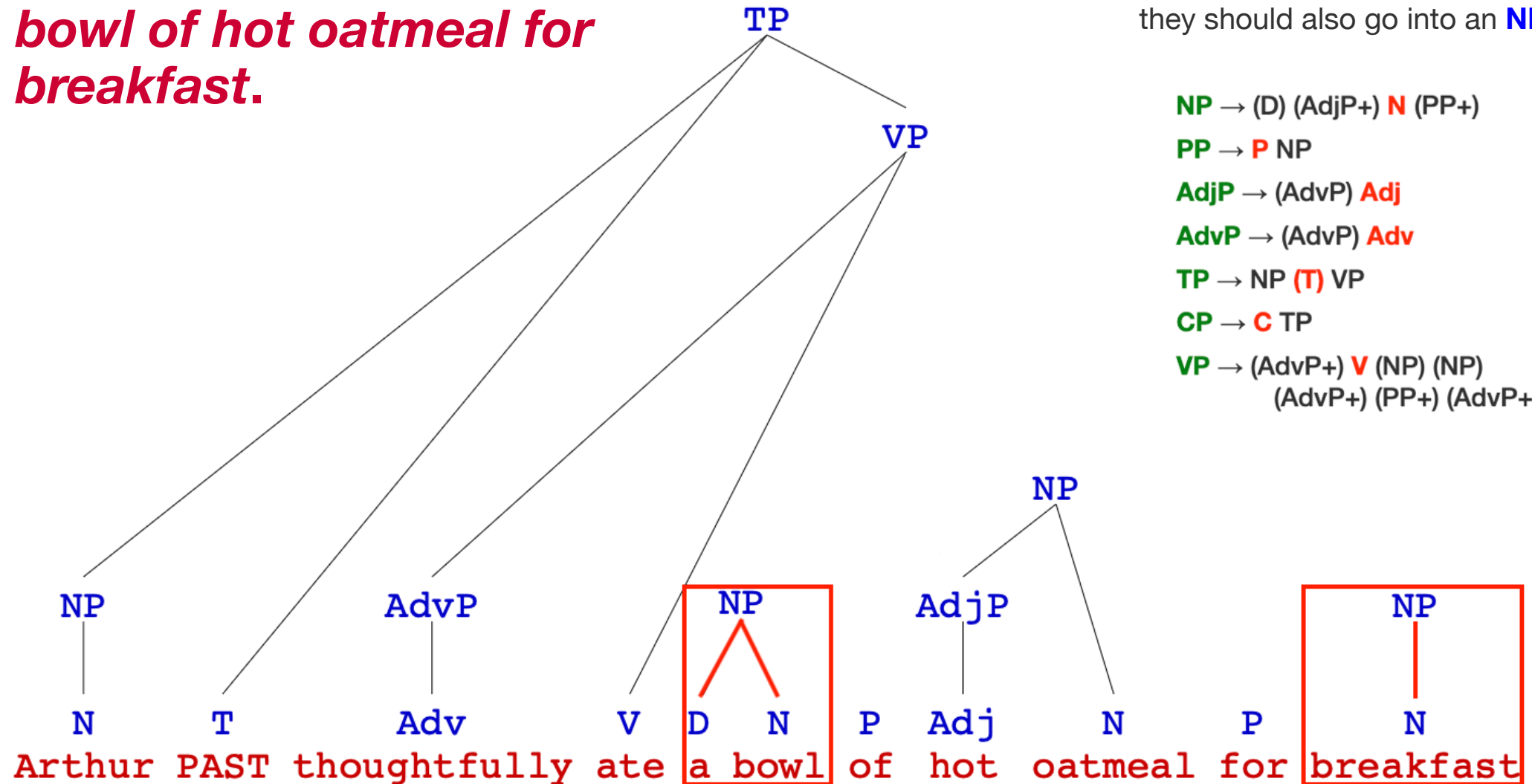


- NP** → (D) (AdjP+) **N** (PP+)
- PP** → **P** NP
- AdjP** → (AdvP) **Adj**
- AdvP** → (AdvP) **Adv**
- TP** → NP (**T**) VP
- CP** → **C** TP
- VP** → (AdvP+) **V** (NP) (NP)
(AdvP+) (PP+) (AdvP+)

Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

if there are any other nouns
they should also go into an **NP**

- NP → (D) (AdjP+) **N** (PP+)
- PP → **P** NP
- AdjP → (AdvP) **Adj**
- AdvP → (AdvP) **Adv**
- TP → NP (**T**) VP
- CP → **C** TP
- VP → (AdvP+) **V** (NP) (NP)
(AdvP+) (PP+) (AdvP+)



Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

are there any **prepositions** you can draw **PPs** for?
any **PP** should contain the **P** and the following **NP**

NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

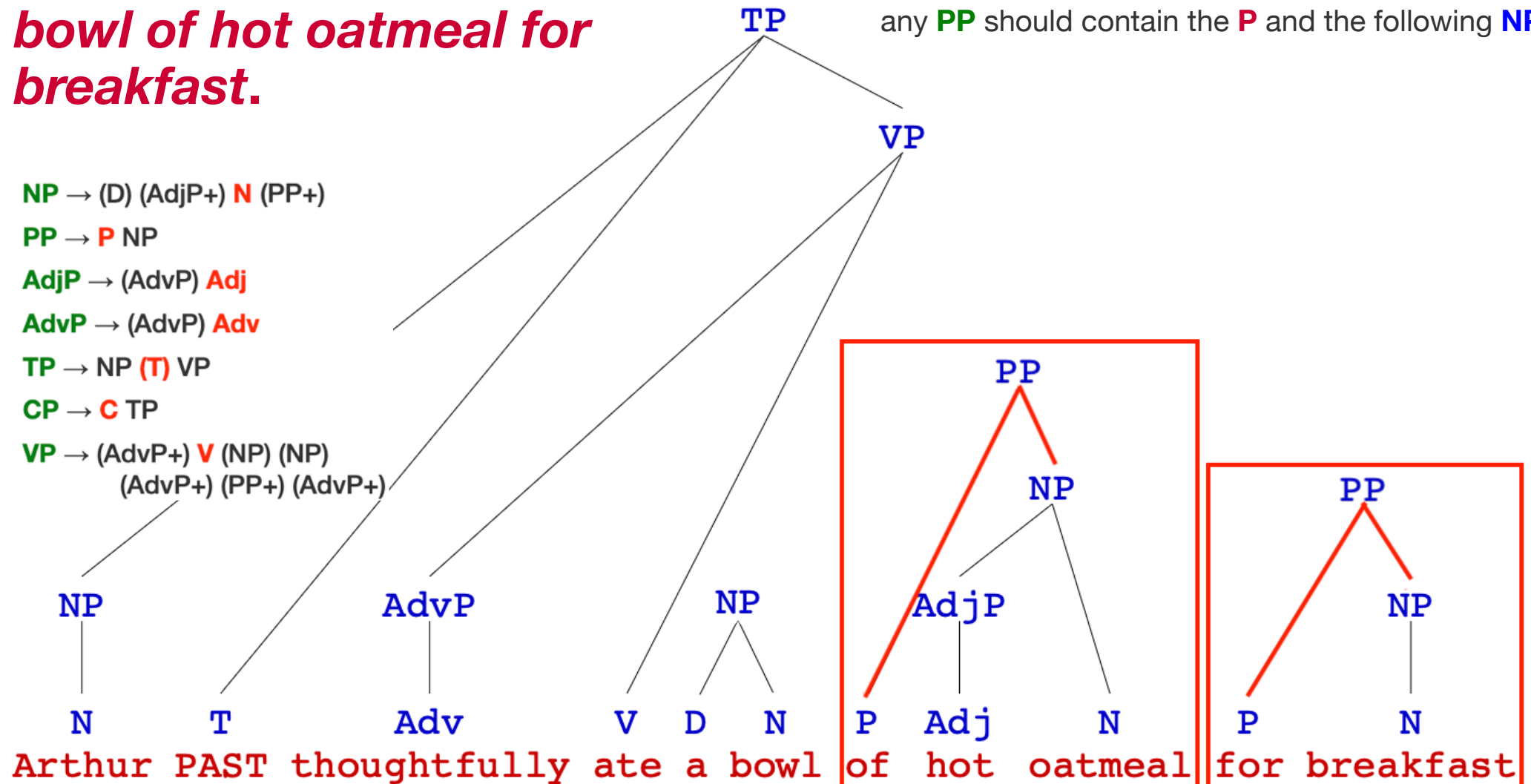
AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

TP → NP (**T**) VP

CP → **C** TP

VP → (AdvP+) **V** (NP) (NP)
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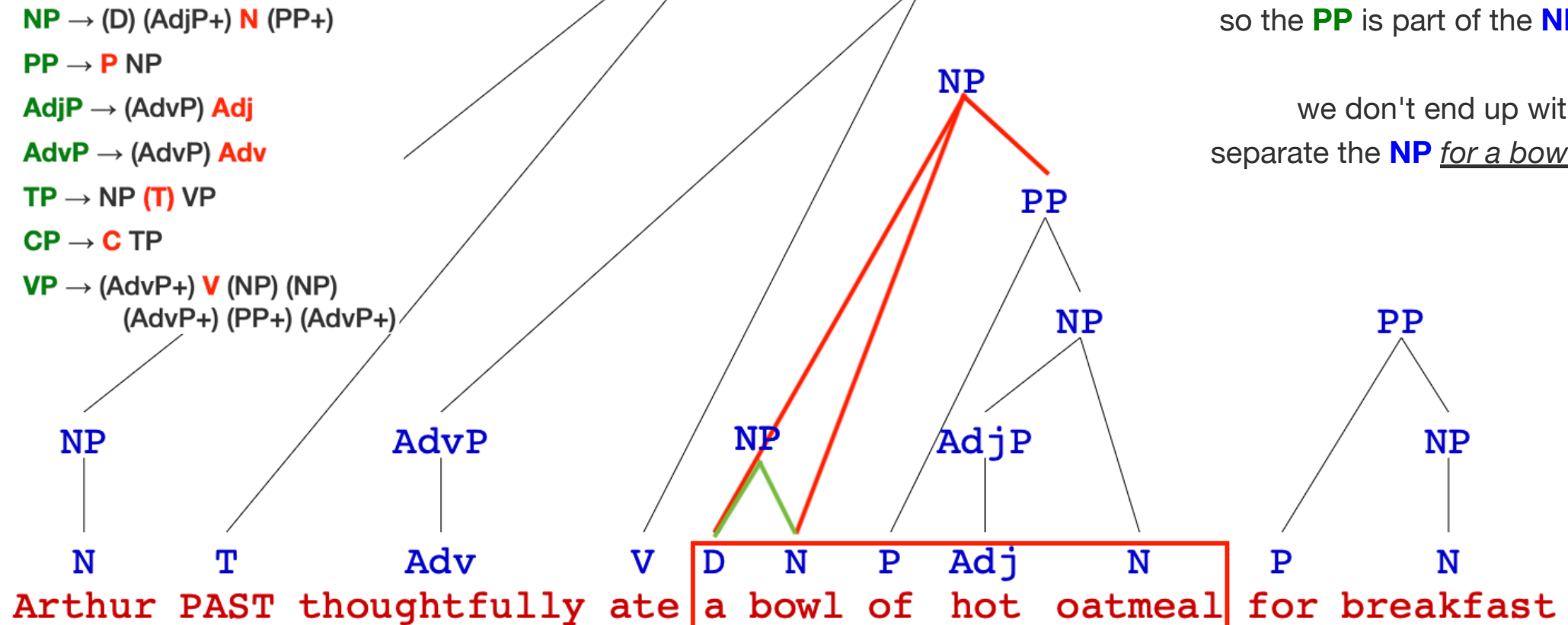


Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

what does the **PP** of hot oatmeal modify?
it needs to be a **sibling** of the word it modifies

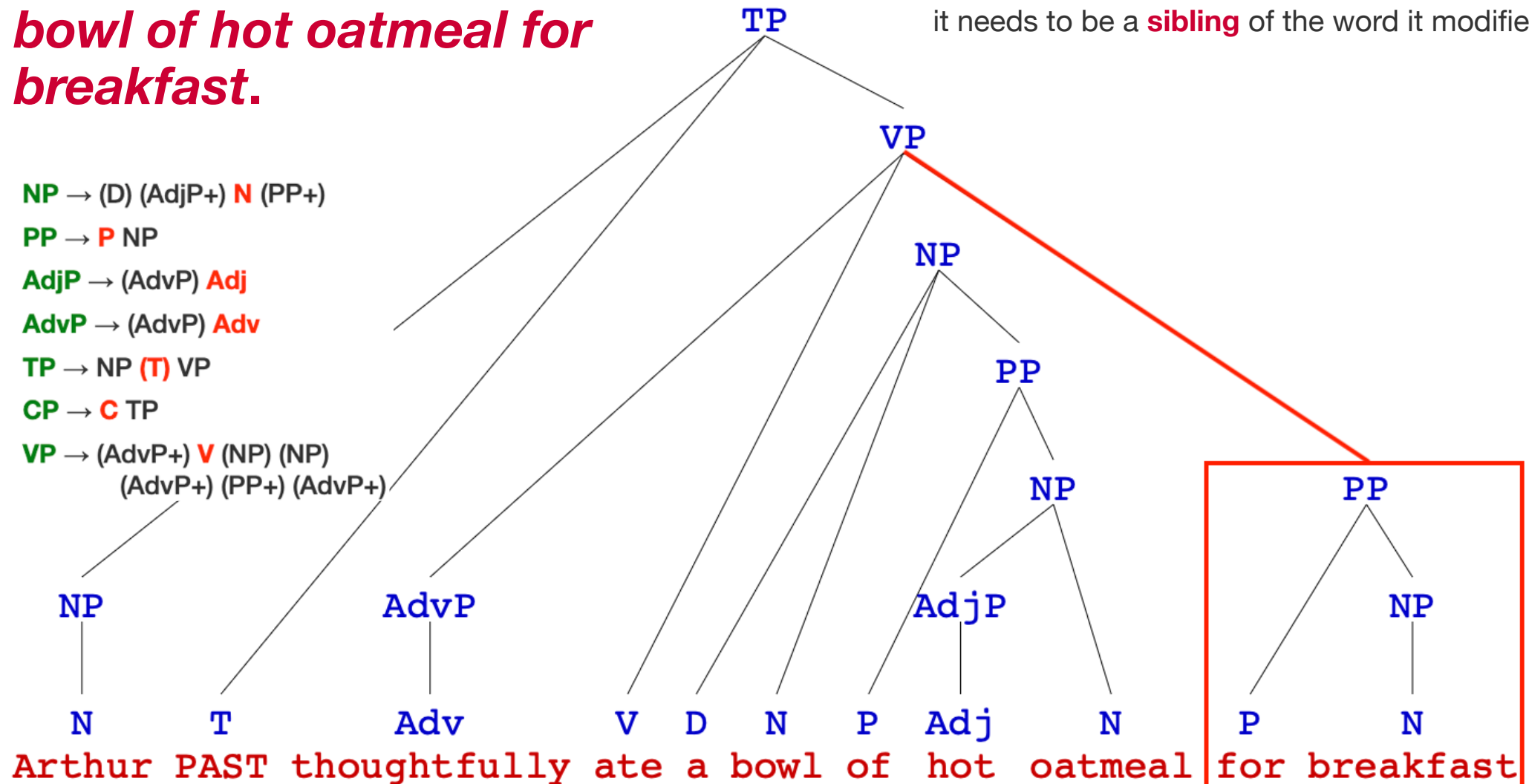
recall that an **NP**
includes everything that modifies **N**
so the **PP** is part of the **NP**

we don't end up with
separate the **NP** for a bowl.



Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

what does the **PP** for breakfast modify?
it needs to be a **sibling** of the word it modifies

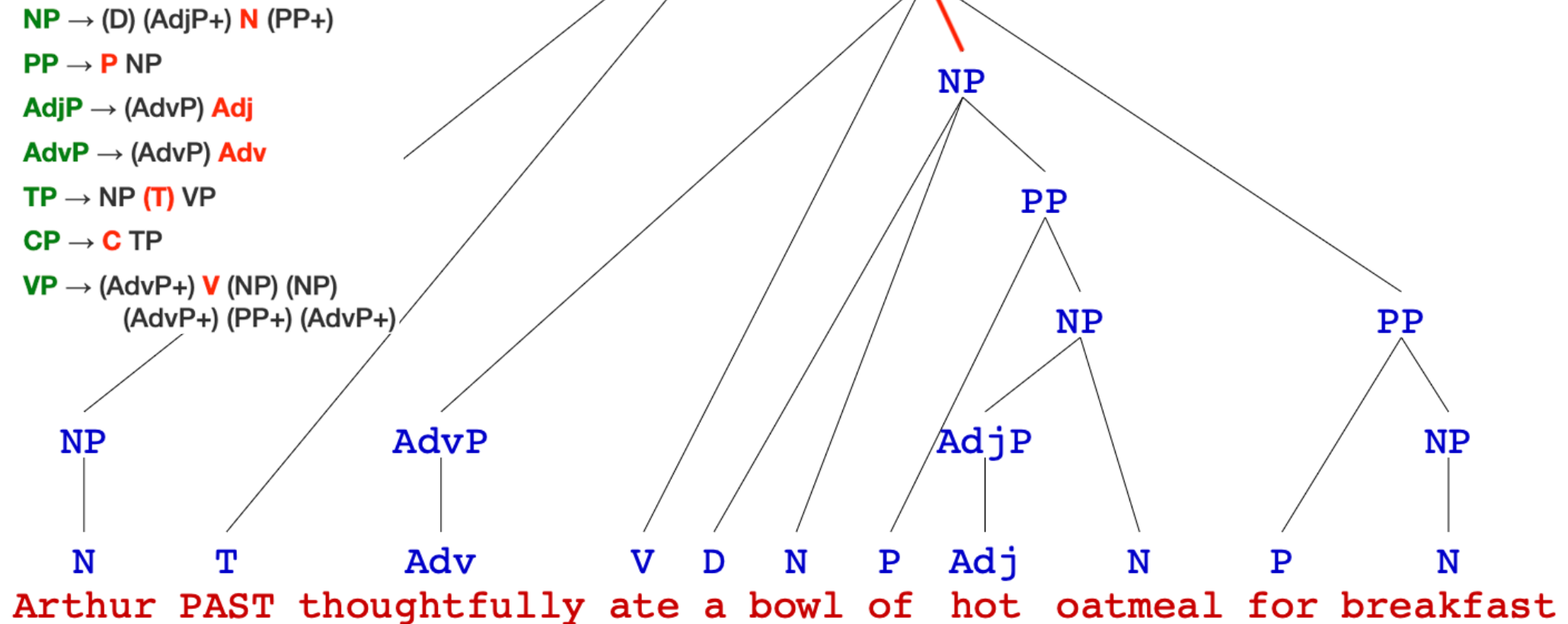


Arthur thoughtfully ate a bowl of hot oatmeal for breakfast.

does the verb ate have any objects?

or what is **eaten**?

that something goes directly under the **VP**



Exercise II

VP, TP

draw **syntax trees** for the following sentences.

- The robin caught a worm.
- The mockingbird was singing.
- The sparrow on the windowsill chirped loudly.
- The osprey brought its hungry chicks a fish.
- The large owl lives in a nest in the barn.
- The duck quickly led her ducklings across the street.

NP → (D) (AdjP+) **N** (PP+)

PP → **P** NP

AdjP → (AdvP) **Adj**

AdvP → (AdvP) **Adv**

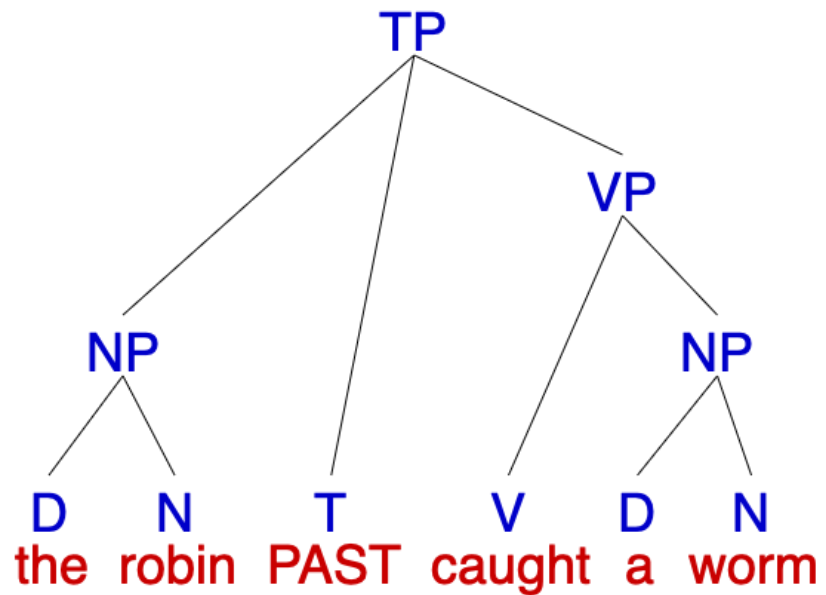
TP → NP (**T**) VP

CP → **C** TP

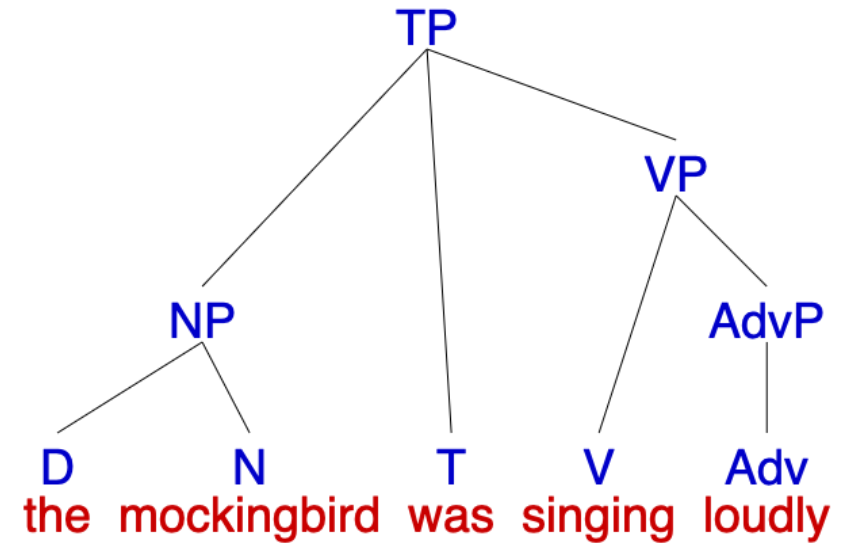
VP → (AdvP+) **V** (NP) (NP) (AdvP+) (PP+) (AdvP+)

Exercise II

VP, TP



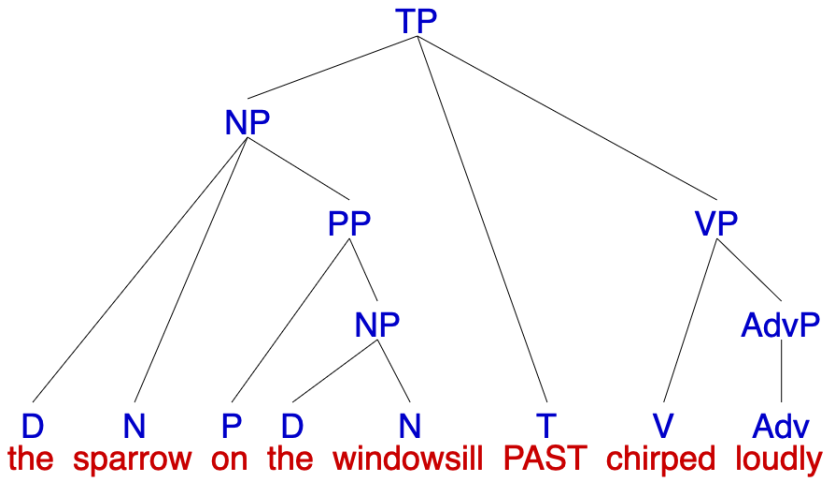
a. 'The robin caught a worm.'



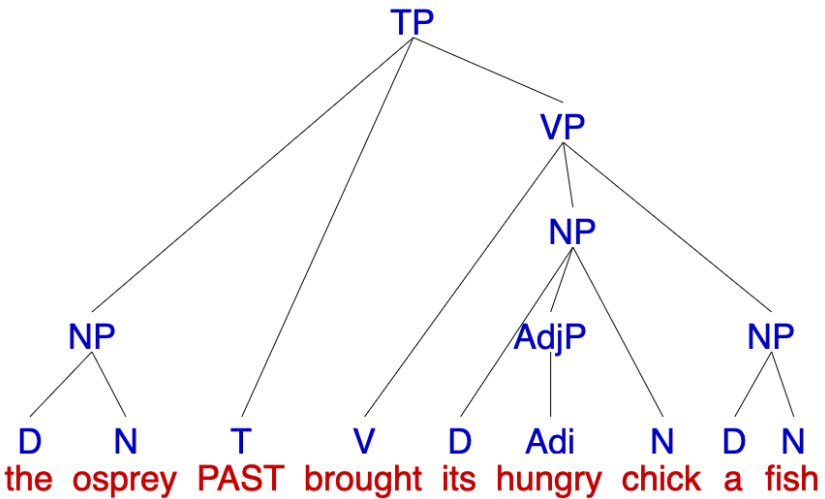
b. 'The mockingbird was singing loudly.'

Exercise II

VP, TP



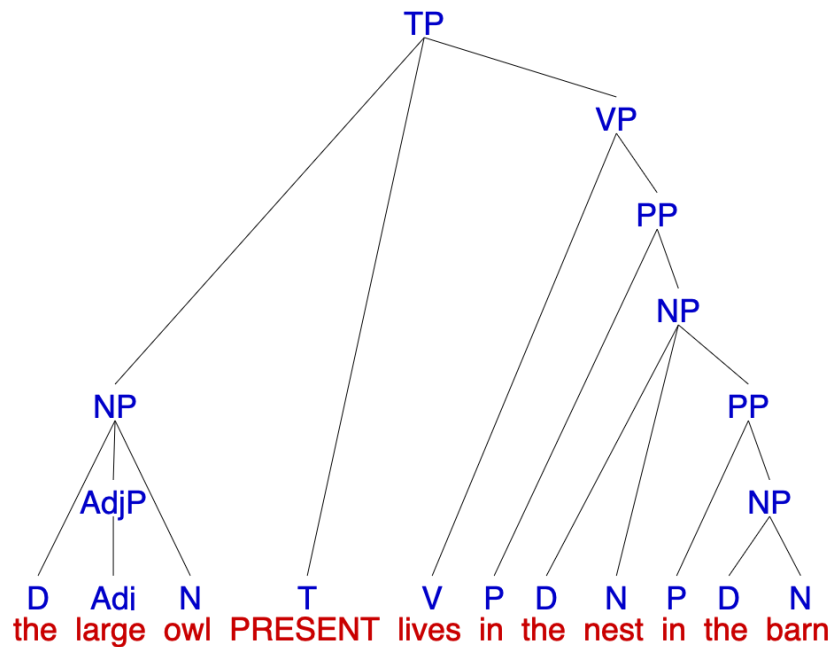
c. 'The sparrow on the windowsill chirped loudly.'



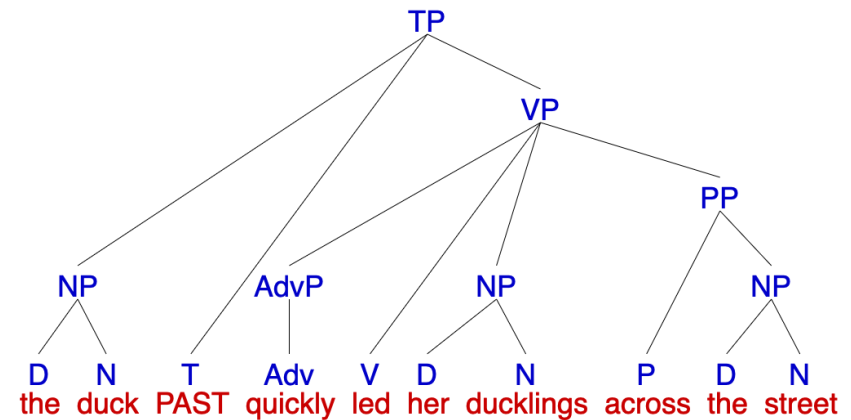
d. 'The osprey brought its hungry chicks a fish.'

Exercise II

VP, TP



e. *'The large owl lives in a nest in the barn.'*



f. 'The duck quickly led her ducklings across the street.'

Assignment III is due this Sunday (**Oct 11**) at **11:59 pm**

continue reading Chapter 3: Constituency, Trees, and Rules from *Andrew Carnie's Syntax textbook*

start preparing for the **MIDTERM**, materials uploaded to the canvas



Slides created via the R package **xaringan**.